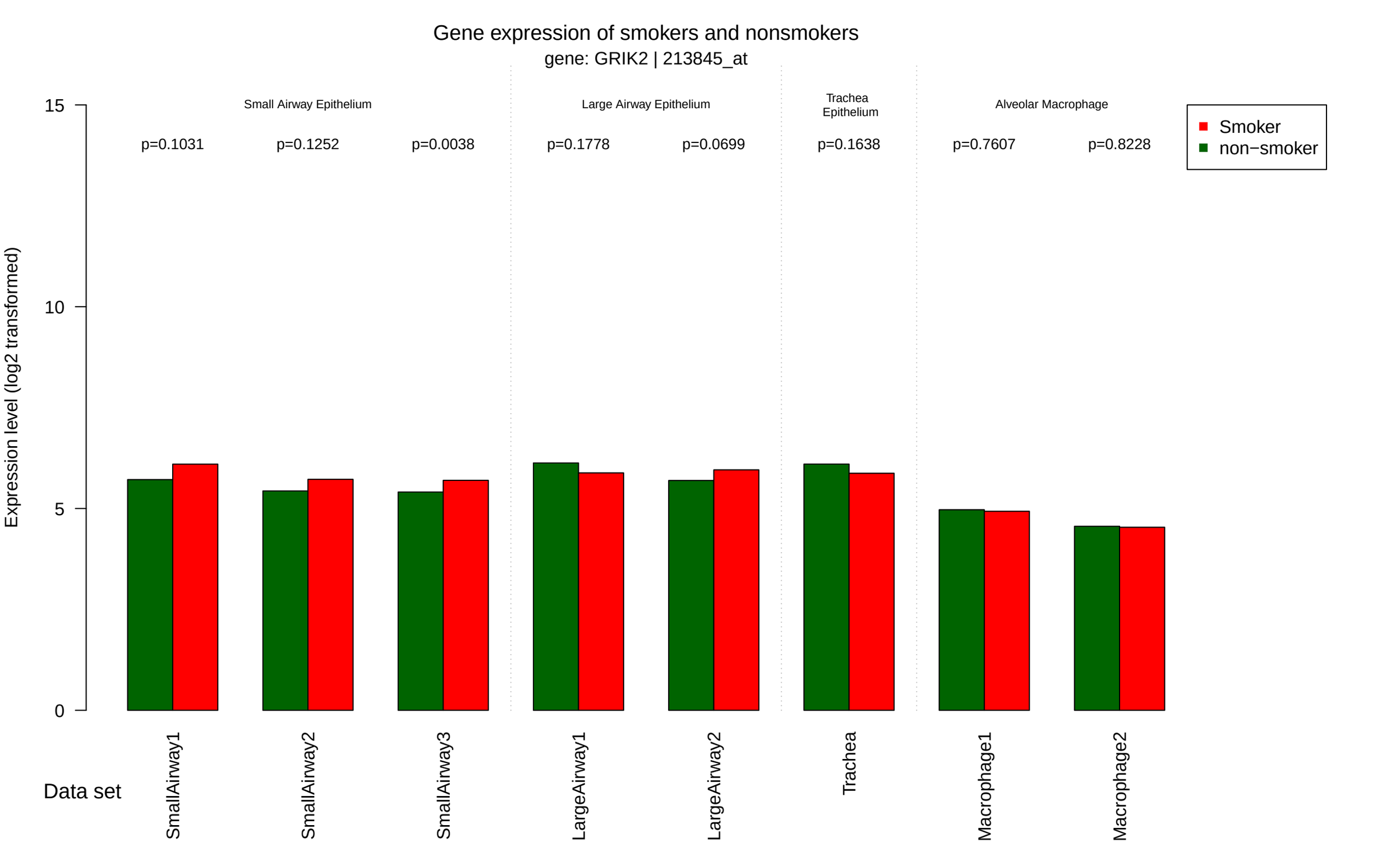
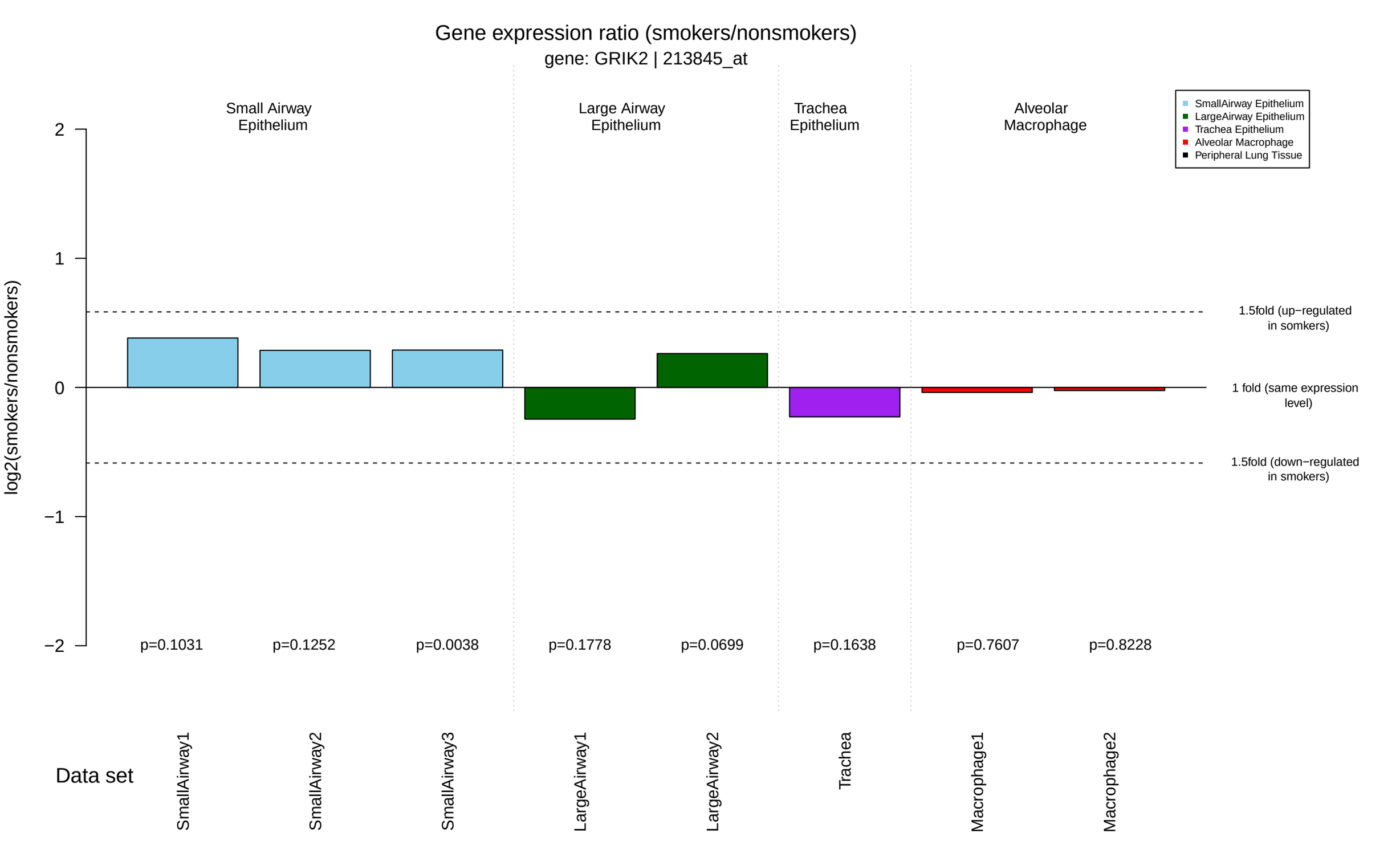
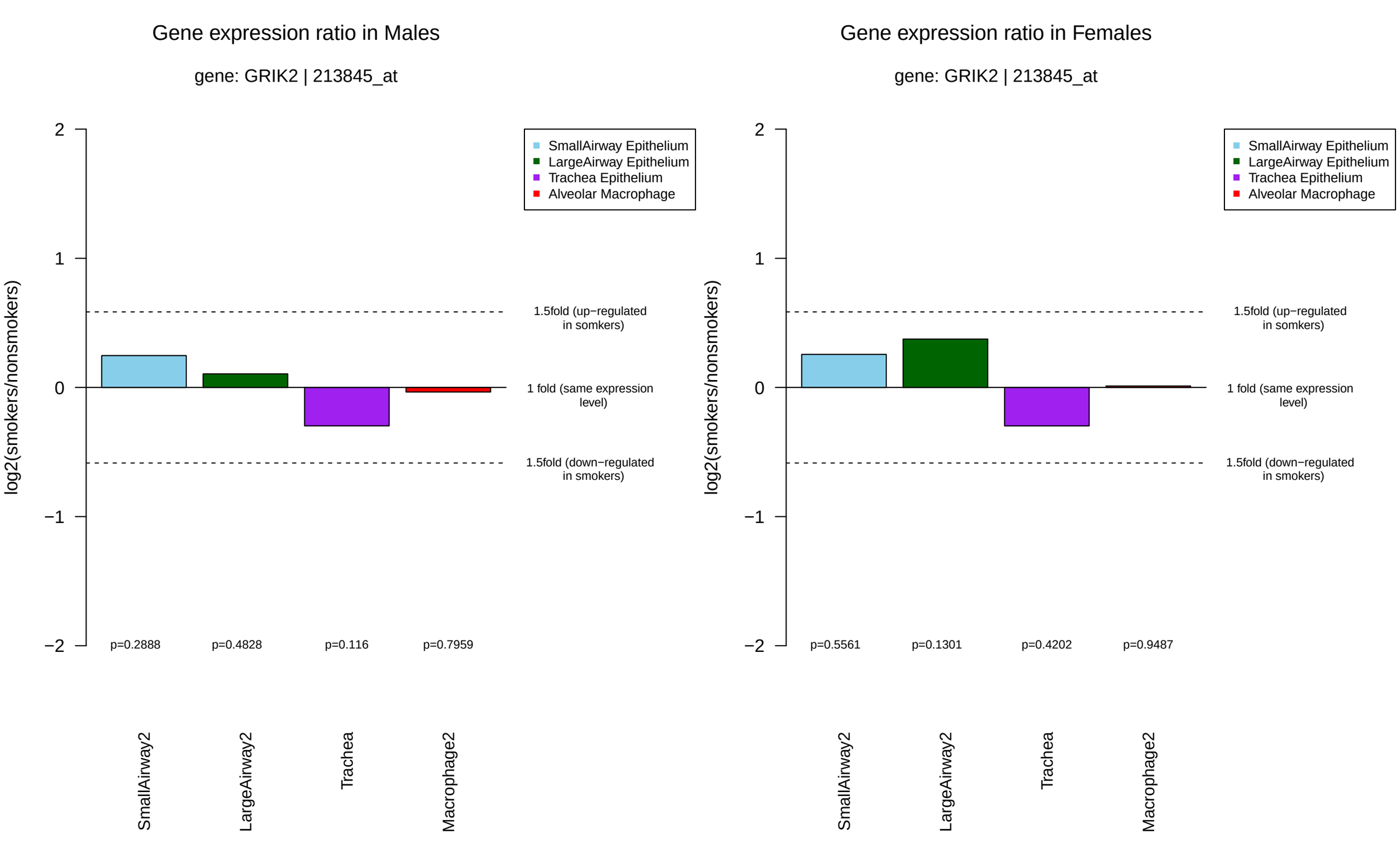
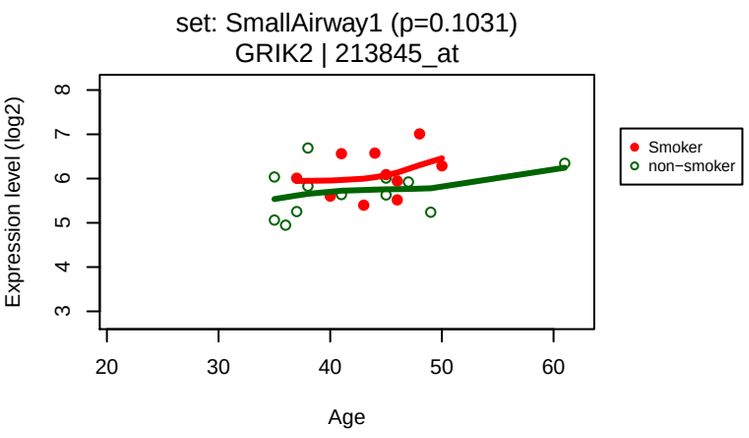
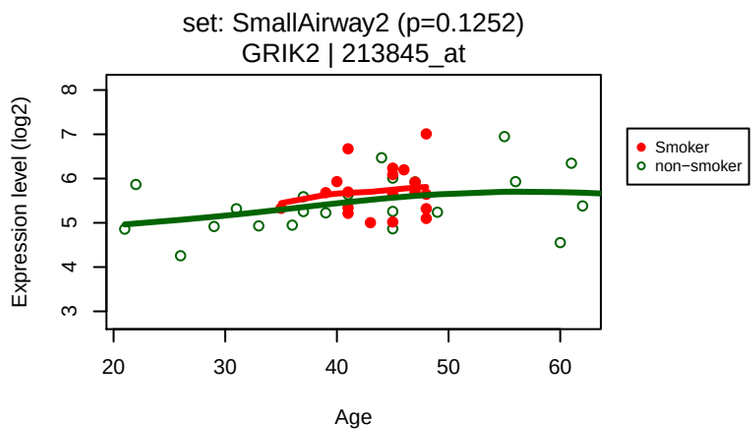
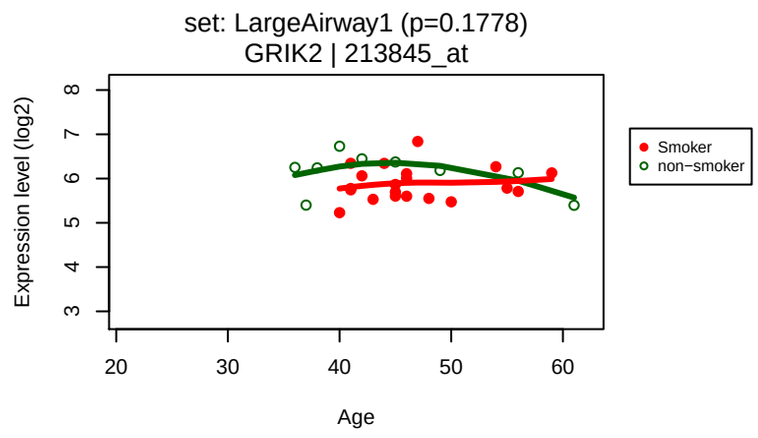

Gene expression of smokers and nonsmokers
gene: GRIK2 | 213845_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1031
p=0.1252
p=0.0038
p=0.1778
p=0.0699
p=0.1638
p=0.7607
p=0.8228
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

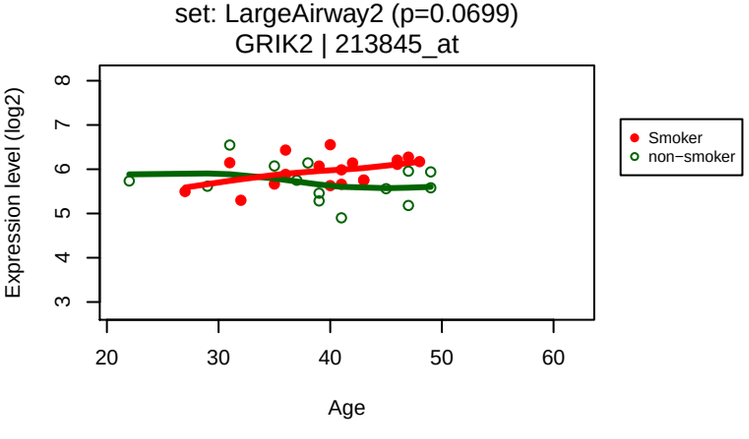

Gene expression ratio (smokers/nonsmokers)
gene: GRIK2 | 213845_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1031
p=0.1252
p=0.0038
p=0.1778
p=0.0699
p=0.1638
p=0.7607
p=0.8228
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

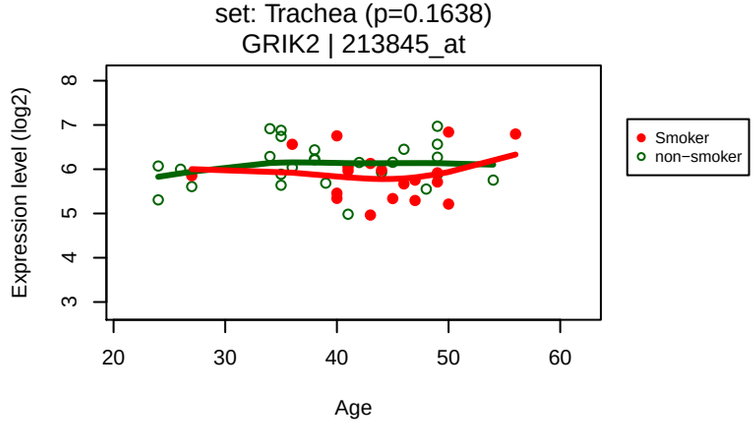

Gene expression ratio in Males
gene: GRIK2 | 213845_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2888
p=0.4828
p=0.116
p=0.7959
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRIK2 | 213845_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5561
p=0.1301
p=0.4202
p=0.9487
SmallAirway2
LargeAirway2
Trachea
Macrophage2

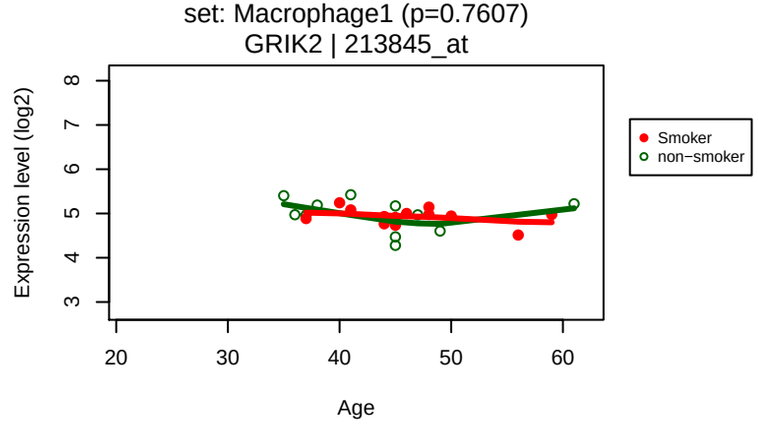

set: SmallAirway1 (p=0.1031)
GRIK2 | 213845_at
Expression level (log2)
Age
Smoker
non−smoker

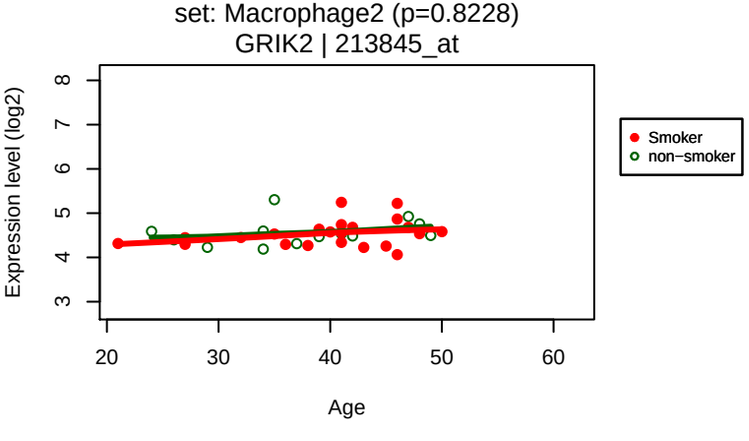

set: SmallAirway2 (p=0.1252)
GRIK2 | 213845_at
Expression level (log2)
Age
Smoker
non−smoker

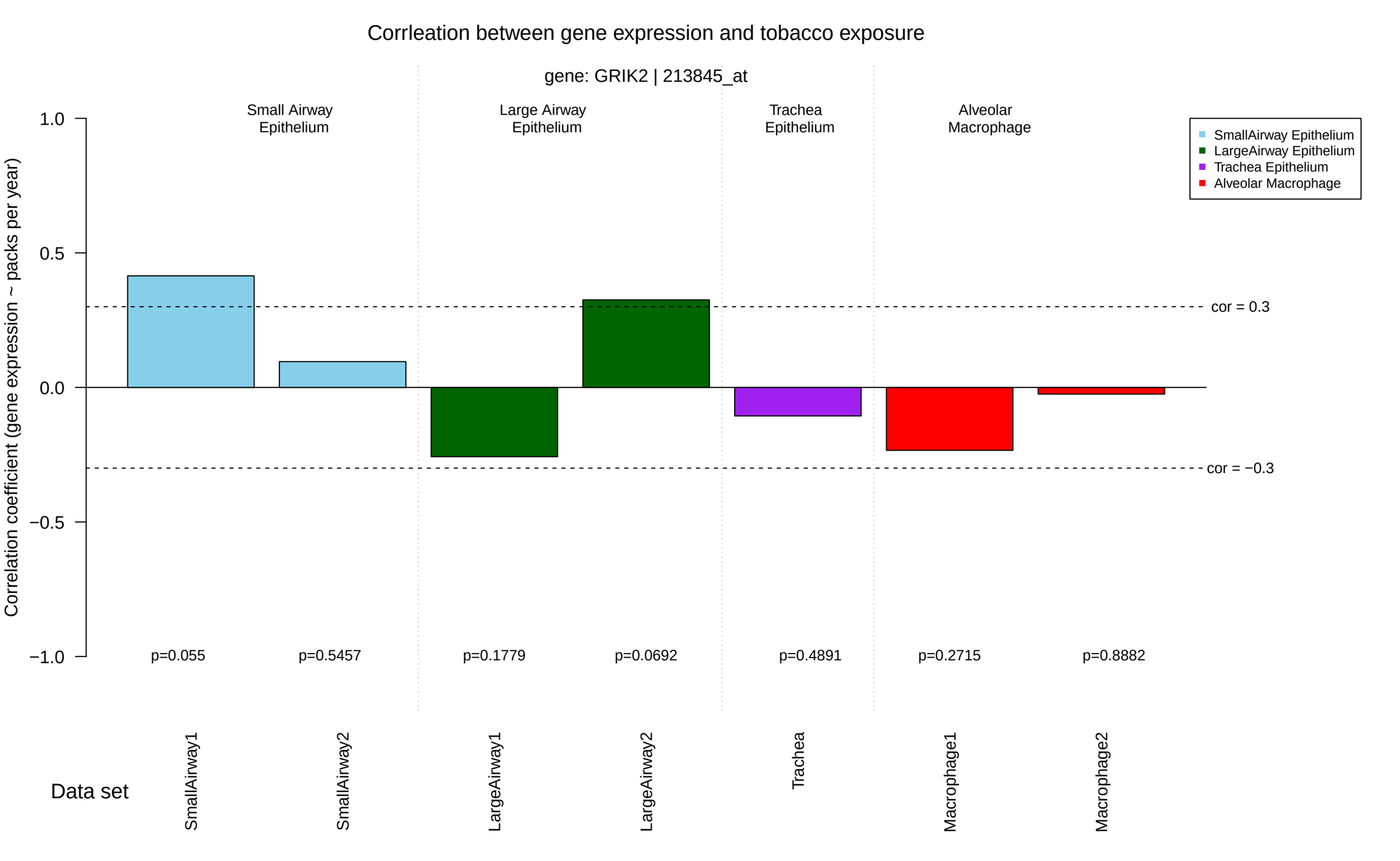

set: LargeAirway1 (p=0.1778)
GRIK2 | 213845_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0699)
GRIK2 | 213845_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1638)
GRIK2 | 213845_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7607)
GRIK2 | 213845_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8228)
GRIK2 | 213845_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRIK2 | 213845_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.055
p=0.5457
p=0.1779
p=0.0692
p=0.4891
p=0.2715
p=0.8882
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2